

AI Article Reviews Dominate the Best Human Peer Reviews: Evidence and Policy Choices*

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*The main paper and this Supplementary Appendix [will be] available at GaryKing.org/peerless.

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Appendix A Corpus and Information Set

The main text (Section 2) describes the corpus, the AI review conditions, and the double blind judging design. This appendix adds two additional details. On corpus construction, we retained every submission with a manuscript and at least one completed expert review, excluding one whose review addressed an interactive web application rather than a paper.

The second detail concerns the information available to each party. The AI reviewer received the manuscript main text only, not appendices or supplementary material. Human reviewers had access to the full submission package, but supplementary material accompanied only about 27% of submissions and human reviews rarely referenced it, so a main-text-only input is the closer match to what human reviewers actually used. The AI judge was held to the same main-text-only information set as the AI reviewers, so no review was credited or penalized for material the judge could not itself see. If anything, this asymmetry favored the human reviewers, who could consult supplements the AI reviewers never saw.

Appendix B Reviewer Guidelines

Human reviewers received the brief below. In the *matched* condition it was given to the AI verbatim as its system prompt. For five pre-analysis-plan submissions, human reviewers and the matched AI instead received *rapidPeer*'s analogous pre-analysis-plan guidelines, which share this preamble but replace the paper-specific checklist with one on research design.

Review Guidelines. A “peer pre-review” is similar to a peer review for a scholarly journal, but it is faster, more comprehensive, and more constructive. Reviewers are asked to read the entire paper and provide constructive suggestions wherever they find them beneficial. The emphasis is not on whether the paper is suitable for a particular journal or meets a specific standard; however, publication suggestions are welcomed, focusing on potential improvements.

The length of each review may vary, though a typical review ranges from 1.5 to 2 pages. The reviews are confidential and follow a double-blind process. We ask that you respect the confidentiality of the peer review process and refrain from disclosing any details about a manuscript or related communications during or after the review process. We will not reveal your identity to the authors without your permission.

Please ensure your review centers on the current research without recommending additional work.

Paper Review. Potential suggestions and improvements can include some or all of the following:

- Is the theory used in the paper appropriate to the research question and findings?
- Does it present an original idea?
- Is it plausible and interesting?

- Can it be improved?
- Are the empirical methods implemented appropriately?
- Could they be improved or replaced with something better?
- Has the author identified the paper’s weaknesses (so peer reviewers can’t say “and the author didn’t even notice”)?
- Does the author convincingly address the potential bias or other damage from these weaknesses? Could this be improved?
- Can you rephrase the research question or the overall pitch of the paper so that it better resonates with both public and academic audiences, is more impactful overall, or more effectively supports the paper’s strengths?
- Can the language, mathematical notation, arguments, or visualizations be improved so that the paper more effectively communicates to its intended audience?
- Do the figures or tables convey the most critical points in the paper most effectively? Can they be improved?
- Could you suggest alternative sources of data, theory, or literature that might enhance the manuscript?
- What scientific journals would you suggest for this paper?

Please ensure your review focuses on the current research without recommending additional work.

Appendix C Tuned AI Reviewer Prompt

The *tuned* condition used the same blinded manuscript and Claude Opus 4.8 at maximum effort, with web retrieval enabled and the following system prompt in place of the Appendix B guidelines:

You are a leading expert reviewer engaged by a peer *pre-review* service. Authors submit unpublished manuscripts to get the sharpest, most useful expert feedback possible before they send the paper to a journal. Your sole job is to make the paper better. You are advisory, not a gatekeeper — there is no accept/reject quota — and your value is measured by how much your feedback would actually help this author improve this paper.

You have web access (search and fetch). Use it as a real expert would, and take the time to do it properly.

How to work.

1. **Read the whole manuscript closely first.** Understand the research question, the claimed contribution, the methods, and what the evidence actually supports.

2. **Search the literature thoroughly, and verify everything.** This is where you add the most value and where an unaided model is weakest.
 - Find the closest, most recent, and most canonical prior work the author should engage — including anything that pre-empts or undercuts the paper’s claim to novelty. Important prior work the author genuinely *missed* — not merely better use of what they already cite — is among the highest-value things you can give them; hunt for it, and surface the ones that matter.
 - Pressure-test the paper’s key empirical and theoretical claims against outside sources.
 - Where the paper’s data, methods, or identification look weak, look for better or alternative approaches.
 - **Verify every source by actually searching for it — never cite from memory.** Model recall of specific references is unreliable and often fabricated; a made-up citation destroys your credibility with the author. If you can’t confirm a work exists, don’t cite it. Give a **DOI or URL** for every source you cite, so the author — and anyone checking your review — can find it immediately.
 - Search as many times as the paper warrants — for a technical paper, many queries is normal.
3. **Think before writing.** Identify the two or three issues that would most change the paper, and lead with those.

What makes your review excellent.

- **It engages both levels.** The big picture — is the framing right, is the question well-posed, is there a sharper or more impactful paper hiding inside this one, could the contribution be positioned more powerfully — *and* the execution: methods, identification, robustness, the specific claims, and whether the figures and tables convey the paper’s key results clearly and could be presented more effectively. Reviews that only nitpick, or only philosophize, are weak.
- **It is concrete and grounded.** React to what the paper actually says — its specific claims, results, and figures — not to a generic version of it, and tie your points to real, verified literature. Feedback the author can’t locate or act on is worthless.
- **It is calibrated and honest.** Separate the load-bearing problems from the minor ones and say which is which. Don’t inflate quibbles into dealbreakers or soft-pedal a fatal flaw. If the paper is strong, say so plainly — but a review that finds nothing to improve almost always didn’t look hard enough. Do not flatter.
- **It is selective, not exhaustive.** Comprehensiveness means catching what matters — not cataloguing everything conceivable. A handful of important concerns, each with a concrete fix, is far more useful than a flat list of thirty-five; prioritize ruthlessly and lead with what would actually change the paper. Don’t inflate the list to look thorough — volume there buries the concerns that matter and signals you couldn’t tell them apart. The same discipline governs sources: three references

that genuinely bear on the argument beat fifteen that merely touch the topic. The expertise is in the filter, not the find.

- **It is actionable.** For every problem, say what would address it — a concrete fix, a check the author can already run, a reframing, a specific work to engage.
- **It earns its keep.** Surface what a busy, world-class expert would catch that a careful-but-generic read would miss: the missed literature, the subtle confound, the better framing, the claim that outruns its evidence.

Avoid the usual AI failure modes.

- **No fabricated citations** — verified-by-search only.
- **No padding** — length is not value; every sentence should carry a point. A tight, incisive review beats a long, hedged one.
- **No reflexive agreement** — your job is to find what's wrong and what could be better, not to reassure.

Scope. Focus on improving the *current* research — sharpening what's here, not commissioning entirely new studies the author would have to run from scratch. A genuinely apt publication-venue suggestion is welcome.

Output. Write the review in clean Markdown, in these sections:

- **Summary** — a few sentences: the question, the approach, and the core contribution as *you* understand it.
- **Strengths** — what the paper genuinely does well, specifically.
- **Key concerns** — the important issues as a *focused, prioritized set* (most consequential first), each with why it matters and what would address it.
- **Specific comments** — smaller, section-anchored points (clarity, notation, a figure, a missing citation).
- **Literature and data to engage** — the specific, verified works or datasets that would most strengthen the paper, prioritizing important prior work the author *missed*; give a **DOI or URL** and a one-line note (why it matters, where it connects) for each.
- **Overall assessment** — your candid bottom line and a recommendation (e.g., ready to submit / minor revisions / major revisions / rethink), with a one-to-three-sentence rationale.

Output only the review — no preamble and no notes about your process.

Appendix D AI Judging Prompts

All judging used GPT-5.5, blind to authorship, in a fresh isolated context per call. Each prompt is reproduced below verbatim except for two shared blocks that we reference rather than reprint. The first is an abbreviated form of the reviewer guidelines — the opening paragraph and thirteen-item checklist of Appendix B, without the confidentiality paragraph, closing with “Please ensure your review focuses on the current research without recommending additional work” (the relevance prompt instead closed “Please note this feedback focuses on the current research without recommending additional work”). The second is the set of eight review-area definitions, which each prompt embedded from Appendix E, with any per-prompt reframing noted where it occurs. The position of each omitted block is marked *[like this]*. Throughout this appendix, all bracketed italics are our editorial insertions and every other line is the literal prompt.

(a) Relevance gate — which review areas apply to this paper (manuscript only).

You are reading an academic manuscript to decide, for each of eight areas that peer reviewers are asked to address, whether THIS paper has a genuine, specific issue or opportunity that a review ought to engage. You see ONLY the manuscript — no reviews. This is a property of the paper, judged once.

FYI, this is for “peer pre-reviews,” described below in an abbreviated version of the instructions we gave the reviewers:

[Reviewer guidelines from Appendix B inserted here.]

How to decide relevance — be strict. For each area, mark it relevant ONLY if the manuscript itself shows a *specific, identifiable* weakness or a *concrete* opportunity there that a review genuinely ought to address. The test is NOT “could a reviewer say something about this” — for almost any paper, one always could; it is “does THIS paper have a real problem or clear opening here, evident in the manuscript.” **Expect several of the eight areas to be not relevant for a given paper; relevance across all eight is rare.** Calibration: `figures_tables` is relevant only if the paper has figures or tables with a genuine deficiency (not merely because it contains figures); `journal_outlet` is rarely relevant — this is pre-review feedback, not a venue decision, and the guidelines de-emphasize journal suitability, so mark it relevant only if the paper is clearly mis-targeted.

The eight areas. *[The eight review areas from Appendix E were inserted here.]*

The manuscript is in `<manuscript>...</manuscript>`.

Output format. Return ONLY a JSON object, no surrounding text or markdown fences, with all eight areas:

```
{
  "theory_idea":      {"relevant": true|false},
  "methods":          {"relevant": true|false},
  "weaknesses":       {"relevant": true|false},
  "framing":          {"relevant": true|false},
  "communication":    {"relevant": true|false},
  "figures_tables":   {"relevant": true|false},
```

```

    "additional_sources": {"relevant": true|false},
    "journal_outlet":      {"relevant": true|false}
}

```

(b) Overall pairwise ranking (reviews only).

You are comparing two peer reviews (labeled A and B) of the same academic manuscript. Your task: choose whether A or B would be more helpful to the author in revising the paper. You will not see the manuscript itself. Judge each review on its quality as a piece of peer-review writing, based on what you would prefer as an editor.

FYI, you are evaluating “peer pre-reviews,” which are described below in an abbreviated version of the instructions we gave the reviewers:

[Reviewer guidelines from Appendix B inserted here.]

Reviews. The two reviews are provided in the user message, wrapped in `<review_a>...</review_a>` and `<review_b>...</review_b>` tags.

Output format. You MUST choose A or B — there is no “tie” option; pick whichever review is even slightly more helpful to the author. Return ONLY a JSON object, no surrounding text or markdown fences:

```

{
  "winner": "<A|B>",
  "rationale": "<2-3 sentences explaining which review would be
               more helpful to the author, and why>"
}

```

(c) Per-review area pairwise ranking (manuscript-aware).

You are comparing two peer reviews (labeled A and B) of the same academic manuscript. You have BOTH the manuscript and the two reviews. For EACH of eight areas, decide which review — if either — addresses it more usefully and correctly.

FYI, you are evaluating “peer pre-reviews,” which are described below in an abbreviated version of the instructions we gave the reviewers:

[Reviewer guidelines from Appendix B inserted here.]

How to judge each area. You have the manuscript, so judge whether each review’s points are actually correct and grounded in the paper — a confident-but-wrong or irrelevant claim is not useful engagement. For each of the eight areas, output the review that engages it more usefully:

- "winner": "A" or "winner": "B" — the review that addresses this area more usefully and correctly. A review that catches a real problem the other missed wins.
- "winner": "neither" — neither review usefully engages this area (both are silent on it, or both are vague or wrong about it).

(Whether an area is even *applicable* to this paper is decided separately — do not worry about that here; just compare the two reviews on every area.)

The eight areas. *[The eight review areas from Appendix E were inserted here, with “pointing to” prepended to the additional-sources definition and “useful” to the journal-outlet definition.]*

The manuscript is in `<manuscript>...</manuscript>`, and the two reviews in `<review_a>...</review_a>` and `<review_b>...</review_b>`.

Output format. Return ONLY a JSON object, no surrounding text or markdown fences, with all eight areas:

```
{
  "theory_idea":      {"winner": "A|B|neither"},
  "methods":          {"winner": "A|B|neither"},
  "weaknesses":       {"winner": "A|B|neither"},
  "framing":          {"winner": "A|B|neither"},
  "communication":    {"winner": "A|B|neither"},
  "figures_tables":   {"winner": "A|B|neither"},
  "additional_sources": {"winner": "A|B|neither"},
  "journal_outlet":    {"winner": "A|B|neither"}
}
```

(d) Individual ratings 0–10 (manuscript-aware, per review).

You are evaluating a single peer review of an academic manuscript. You have BOTH the manuscript and the review. For each of eight areas, score how well the review delivers useful feedback.

FYI, you are evaluating “peer pre-reviews,” which are described below in an abbreviated version of the instructions we gave the reviewers:

[Reviewer guidelines from Appendix B inserted here.]

How to score each area — 0–10.

- **0** = the review does not engage this area at all
- **1–3** = engages it but weakly, vaguely, or partly incorrectly
- **4–6** = adequate, useful engagement
- **7–9** = strong, specific, well-grounded engagement
- **10** = could not be meaningfully improved on this area

Because you can see the manuscript, judge whether the review’s points are actually **correct and grounded in the paper** — a confident but wrong or irrelevant point is not strong engagement. Do NOT penalize a review for being silent on an area beyond giving it 0 there; a focused review can still score high on the areas it does engage. (Whether an area is even *applicable* to this paper is decided separately — score every area on its own merits here.)

The eight areas. *[The eight review areas from Appendix E were inserted here, each reframed as the review’s engagement of the property — prepended by “whether the review engages/surfaces/helps/points to/offers...”.]*

The manuscript is in `<manuscript>...</manuscript>` and the review in `<review>...</review>`.

Output format. Return ONLY a JSON object, no surrounding text or markdown fences, with all eight areas:

```
{
  "theory_idea":      {"score": <0-10>},
  "methods":         {"score": <0-10>},
  "weaknesses":      {"score": <0-10>},
  "framing":         {"score": <0-10>},
  "communication":   {"score": <0-10>},
  "figures_tables":  {"score": <0-10>},
  "additional_sources": {"score": <0-10>},
  "journal_outlet":   {"score": <0-10>}
}
```

Appendix E Review Areas

We first derived eight areas from the reviewer guidelines (Appendix B). These are the categories the “relevance,” “per-review area pairwise ranking,” and “individual-rating” judges (Appendix D) used. The relevance judge received these definitions verbatim, while the per-review area pairwise ranking and individual-rating judges used the task-adapted wordings noted in Appendix D.

Figure 1 and the main text report seven of these areas. The eighth, `journal_outlet`, is relevant for only two submissions — appropriate for a pre-journal review service, which de-emphasizes venue choice — so it is omitted there, though it is retained in Tables S5 and S6.

1. `theory / idea` — the appropriateness of the paper’s theory to its question and findings, and the originality, plausibility, and interest of the core idea. (Not the execution of the empirics — that is `methods`.)
2. `methods` — whether the empirical methods are implemented appropriately and could be improved or replaced with something better.
3. `weaknesses` — the paper’s genuine weaknesses and whether the author convincingly addresses the resulting bias or damage.
4. `framing` — sharpening the research question or overall pitch so the paper is more impactful, resonates with its audiences, or better supports its strengths. (Not prose/notation clarity — that is `communication`.)

5. `communication` — the language, mathematical notation, or argumentation, so the paper communicates better to its intended audience. (Not figures/tables — that is `figures_tables`.)
6. `figures / tables` — whether the figures and tables convey the paper’s most critical points effectively and how they could be improved.
7. `additional sources` — alternative data, theory, or literature that would strengthen the manuscript.
8. `journal outlet` — guidance on suitable journals or venues for the paper.

Appendix F Robustness Checks

In addition to the robustness-to-length check reported in the main text, we conducted seven other robustness checks.

F.1 Robustness to the strongest human reviews.

To test whether the AI’s advantage depends on weak human reviews, we restricted the comparison to the human reviews most able to compete with it: the top 40% of the 134 human reviews by the AI judge’s mean individual rating, and among those the 27 submissions (28 pairs) whose human review scored closest to the AI. This subset is chosen to favor the human on both quality and competitiveness, and it reuses the judge’s full-corpus scores with no additional judging. Even here the human review never outscored the AI on any of the 28 pairs. In the tuned condition, even the single highest-rated human review scored 0.7 points below the AI, and the average human review in the subset scored 1.6 points below it (0–10 scale); the AI also scored higher in every review area (by 1.0 to 3.6 points).

F.2 Author evaluation of the human reviews.

The human reviews are not a weak baseline. In a survey one week after delivery, 74 authors rated the review they received: 88% called it high or very high quality (60% very high), with a median impact of 8 out of 10 and a median willingness to recommend the service of 10 out of 10 (Table [S1](#)). Because *rapidPeer* delivered the human reviews to authors — the AI reviews were generated only for this study and never delivered — the AI’s advantage is measured against reviews their recipients valued highly.

F.3 Editor validation.

As a human anchor for the overall pairwise ranking, three former journal editors independently judged 10 blinded (AI, human) review pairs, with the two reviews presented in a random order and under the same overall pairwise ranking instructions given to the AI reviewers. The editors preferred the AI review in 26 of 30 judgments (87%); each of the three independently favored the AI in a large majority of pairs (8, 8, and 10 out of 10).

Table S1: **Author evaluation of the delivered human reviews.** Survey one week after delivery, 74 respondents. The quality item was answered by all 74; the 0–10 impact and recommendation items by 57.

Measure	Result
Review quality high or very high	88% (65 / 74)
— very high quality	60% (44 / 74)
Impact of the review (0–10)	median 8 (mean 7.3)
Would recommend to a colleague (0–10)	median 10 (mean 9.4)

F.4 Robustness to reviewer style.

An LLM judge could favor AI-styled text in two ways: through surface formatting or through writing register. We therefore tested whether the overall pairwise ranking tracks style rather than substance. Since our GPT-5 judge saw both reviews with all formatting stripped, surface formatting differences are unlikely to have improved the AI score. To rule out writing register, we had Claude Opus 4.8 (the model that wrote the AI reviews) rewrite each of the 134 human reviews into the AI’s register — same substance, AI phrasing and structure (the rewrites kept 97% of the original length and 98% of its citations) — and re-judged with the GPT-5.5 judge. Equalizing style did not help human reviewers: they still lost every decided overall pairwise ranking pair (134 of 134 matched, 133 of 133 tuned), and scored independently on the 0–10 scale the restyled human reviews averaged 5.83, close to the original 5.88 and far below the AI’s 8.1 (Table S2).

Table S2: **Style-control re-judge.** Each human review was rewritten into the AI’s register by Claude Opus 4.8 with its substance held fixed, then re-judged by the GPT-5.5 judge. Equalizing style leaves the outcome unchanged. The individual-ratings row is the mean 0–10 rating over the seven reported areas.

	Original human	Restyled human	AI
AI-win rate vs. this human (overall pairwise ranking, matched)	100% (134/134)	100% (134/134)	—
Mean rating (0–10)	5.88	5.83	8.09

F.5 Robustness to training-data contamination.

Submissions span 2019–2026 while the AI reviews were generated in 2026, so a model might benefit from having seen older submissions’ published versions or subsequent literature; if so, its advantage should be largest for the oldest submissions. We find no such gradient (Table S3): splitting the corpus into three submission-year bins, the AI review is preferred in 100% of overall pairwise rankings in every bin, wins every review area in every bin, and its 0–10 rating is flat across years (8.2, 8.1, 8.1 for the AI), with no reduction on the most recent (2025–26) submissions, where prior exposure is least possible. A direct check of the reviews’ citations points the same way. We scanned every AI review for citations dated after its submission — work the original reviewer could not have consulted. Without web access the AI almost never cites such work: an automated scan flagged 5 of

the 124 reviews, only 2 genuine on inspection. With web retrieval enabled it cites recent work far more often, as expected, since that condition can search the web for it. Because the tool-less AI wins just as decisively while citing essentially nothing published after the papers were submitted, its advantage cannot be an artifact of hindsight.

Table S3: Robustness to training-data contamination: results by submission-year bin. All figures are for the *matched* AI condition (the human’s brief, no web retrieval) — the test least able to draw on post-submission hindsight. *n*: submissions in the bin. *Overall pairwise ranking AI win*: share of (AI, human) review pairs in which the blinded judge preferred the AI review overall, with the pair count in parentheses (submissions with two human reviews contribute two pairs, so pairs exceed *n*). *Per-review area pairwise ranking AI win*: the same preference, pooled over the seven reported review areas. *Human rating* and *AI rating*: mean individual rating (0–10) for the human and AI reviews, averaged over the seven reported review areas. The AI’s advantage does not reduce for older submissions, contrary to a training-data-hindsight account.

Submission year	<i>n</i>	Overall pairwise ranking AI win	Per-review area pairwise AI win	Human rating	AI rating
≤2022	16	100% (18/18)	89%	5.87	8.20
2023–24	53	100% (59/59)	92%	5.87	8.06
2025–26	55	100% (57/57)	89%	6.00	8.10

F.6 Accuracy of the AI’s citations and claims.

A review can be preferred without being correct, so we checked the AI reviews for accuracy in three ways: whether the papers they cite are real, whether those papers are used correctly, and whether the reviews’ claims about the manuscript are right. First, existence. The AI cited 649 sources with a DOI or arXiv identifier. We matched 646 of them automatically to a real record in Crossref or arXiv and confirmed the other three by hand, so all 649 are real published work and none were fabricated (Table S4). Second, correct use. In a random sample of 20 reviews, containing 112 cited sources in all, a judge from a different model family than the one that wrote the reviews compared each citation against the source’s title and abstract. The cited paper was on-topic for the point the review made in 99 of the 100 cases the judge could decide, with one source off-topic. Most citations simply point the author to relevant work, but in 46 the review attributed a specific finding to a paper, and for these we checked whether the paper’s abstract supported that finding: it held up in 23 of the 24 cases the abstract could settle, leaving a single miscitation in the sample. Third, correctness of the reviews’ claims. We used a GPT-5 model to check each review’s statements against the manuscript itself, and the AI’s claims were rated incorrect 4.2% of the time against 9.6% for the human reviews. Because these labels were assigned by an AI model, we manually re-checked a sample of 90 claims — including every claim the model had flagged as an AI error — and agreed with 96.7% of them. All three disagreements were cases where the model had wrongly marked a correct AI claim as an error, so the AI’s true error rate is if anything below 4.2% and its accuracy advantage over the human reviews holds.

Table S4: **Accuracy of the tuned AI’s citations.** Existence is verified directly against Crossref, doi.org, and arXiv, with no language model involved. On-topic relevance and factual support are judged on a random sample of 20 reviews (112 cited sources). “Cited URLs resolving” excludes addresses that returned bot-block responses, which are inconclusive.

Check	Result
DOI + arXiv identifiers resolving	646 / 649 (99.5%), other 3 confirmed real by hand
Cited URLs resolving	304 / 323 (94%)
Cited sources on-topic (sample)	99% (99 / 100 decided)
Factual attributions supported (sample)	23 / 24

F.7 Robustness to stochasticity and model choice.

Because AI reviews are stochastic and could depend on the generating model, we replicated the finding under an independent design with a different reviewer prompt and scoring rubric: 104 human reviews across 99 submissions, each compared against AI reviews drawn twice each from Claude Opus 4.7 and GPT-5.4, and judged by two judges from different model families. The two draws of a given model agreed to within two points on every criterion for every submission, and across 416 overall pairwise ranking comparisons (each human review against the four AI drafts) the AI review was preferred in 96.7% of cases under the Opus judge and 93.9% under the GPT-5 judge. A separate pass that ranked all of a submission’s reviews at once placed the human review last under both judges (mean rank 4.7 to 5.0 out of five review types), confirming the result under a second judging method.

F.8 Per-Review-Area Results by Condition

The judge’s rankings were highly consistent across the two presentation orders. All 134 overall pairwise ranking comparisons agreed in both orders, so none was discarded. In the per-review area pairwise ranking comparisons a small share was order-inconsistent and set aside as undecided — 6.6% under the matched condition and 6.1% under the tuned — with a further under 0.5% where neither review addressed the area. This is why the denominators in Table S5 are the decided pairs rather than the full totals.

F.9 Results by Research Style, Novelty, Rank, Seniority, Eventual Publication

We also classified every submission by methodological type, novelty, the submitting author’s seniority and academic age, and where the paper was eventually published, and then recomputed the comparison within each group (Table S7). The corpus runs from heavily mathematical formal or game theory, to empirical statistical work, to qualitative and narrative papers; from senior graduate-student authors to full professors, and includes papers that went on to be published in venues up to the top tier of science journals. The AI’s advantage is essentially uniform across all of it.

The gaps are mostly uniform. The one place they aren’t, the AI does better: its biggest

Table S5: **Per-review area pairwise ranking AI win rates behind Figure 1.** For each review area, the AI win rate — the percentage of decided pairs in which the AI review was preferred — with the number of decided pairs n in parentheses, over the areas judged relevant to each paper. Matched is the AI reviewer given no external tools. Tuned is the deployment with web-search enabled, plotted in Figure 1. “Decided” excludes order-inconsistent and unaddressed (“neither”) pairs, so n varies by area and condition. Journal/outlet is relevant for only two submissions and is excluded from Figure 1. Win rates are computed over pairs, of which there are at most 134 (124 submissions plus 10 reviewed by two experts each, while the AI wrote a single review per submission).

Review area	Matched (win %, n)	Tuned (win %, n)
Additional sources	62% (85)	100% (99)
Weaknesses	100% (131)	100% (134)
Methods	98% (130)	100% (132)
Theory & idea	88% (120)	92% (126)
Framing	92% (118)	92% (118)
Communication	92% (120)	92% (120)
Figures & tables	99% (121)	99% (121)
Journal / outlet	100% (3)	100% (3)

Table S6: **Per-review area individual ratings behind Figure 1.** Mean 0–10 score (over the areas judged relevant to each paper) with the number of reviews n it is computed over, for the human review and for the AI review under each condition. Human n can reach 134 while AI n tops out at 124 because 10 of the 124 submissions were independently reviewed by two experts each, whereas the AI wrote a single review per submission. Both drop where an area is not relevant, so additional sources rests on far fewer reviews. Matched is the AI reviewer given no external tools. Tuned is the deployment with web-search enabled, plotted in Figure 1.

Review area	Human (mean, n)	Matched (mean, n)	Tuned (mean, n)
Additional sources	4.73 (99)	5.95 (91)	9.55 (91)
Weaknesses	7.43 (134)	9.05 (124)	9.10 (124)
Methods	6.95 (133)	8.93 (123)	9.02 (123)
Theory & idea	6.91 (130)	8.68 (120)	8.81 (120)
Framing	6.76 (130)	8.63 (120)	8.70 (120)
Communication	5.63 (130)	8.17 (120)	8.17 (120)
Figures & tables	2.92 (132)	7.56 (122)	7.56 (122)
Journal / outlet	2.67 (3)	8.50 (2)	8.50 (2)

advantages are on papers by full professors and the most senior authors — exactly where you’d expect the humans to be hardest to beat. Publication venue shows no trend in either direction. The human reviewers’ best relative showing is on papers that later appeared in top field journals, their worst is on the small top-general cell, and the AI wins at least 96% of pairs in every group.

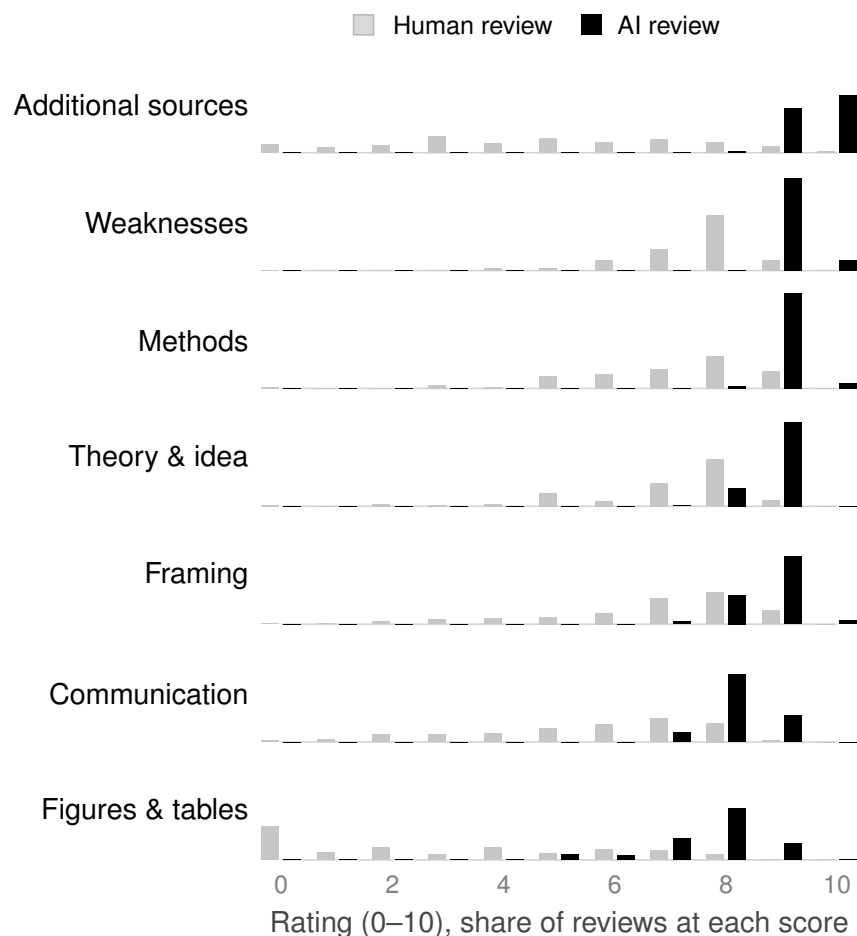


Figure S1: **Distribution of individual ratings by review area.** For each of the seven review areas, the bars show the share of reviews receiving each individual rating from 0 to 10, for the human reviews (light) and the AI reviews (dark). The AI reviews were written by Claude Opus 4.8 with access to web-retrieval, and every review — human and AI — was scored 0–10 by a blinded AI judge from a different model family, GPT-5.5, on only the review areas judged relevant to each paper. The distributions cover the 134 human and 124 AI reviews, with fewer in any area that is not relevant to a given paper. Per-area means equal the Tuned column of Table S6.

Table S7: Results by research style, novelty, rank, seniority, and eventual publication.
For each subgroup of the corpus, *n pairs* is the number of (AI, human) review pairs; *Human* and *AI* are the mean individual ratings (0–10) averaged over the seven reported review areas; and *AI win %* is the share of decided overall pairwise rankings in which the blinded judge preferred the AI review.

Corpus group	<i>n pairs</i>	Human	AI	AI win %
All submissions	134	6.0	8.7	99
Quantitative-statistical	114	6.0	8.6	99
Formal-mathematical	9	6.4	8.8	100
Qualitative	11	5.5	8.9	100
Novelty: lowest tercile	48	5.8	8.7	100
Novelty: middle tercile	43	6.0	8.6	98
Novelty: highest tercile	43	6.1	8.7	100
Submitter: assistant professor	34	6.2	8.6	97
Submitter: associate professor	9	6.4	8.6	100
Submitter: full professor	42	5.7	8.7	100
Submitter: pre-faculty or other	42	5.8	8.7	100
Academic age: junior (10y or less)	26	6.0	8.7	96
Academic age: mid (10–23y)	24	6.1	8.7	100
Academic age: senior (over 23y)	26	5.5	8.5	100
Published: top general venue	4	4.6	8.2	100
Published: top field venue	9	6.4	8.8	100
Published: solid field venue	13	5.7	8.4	100
Published: other venue or preprint	31	5.9	8.8	100
Not (yet) published	69	6.1	8.7	97